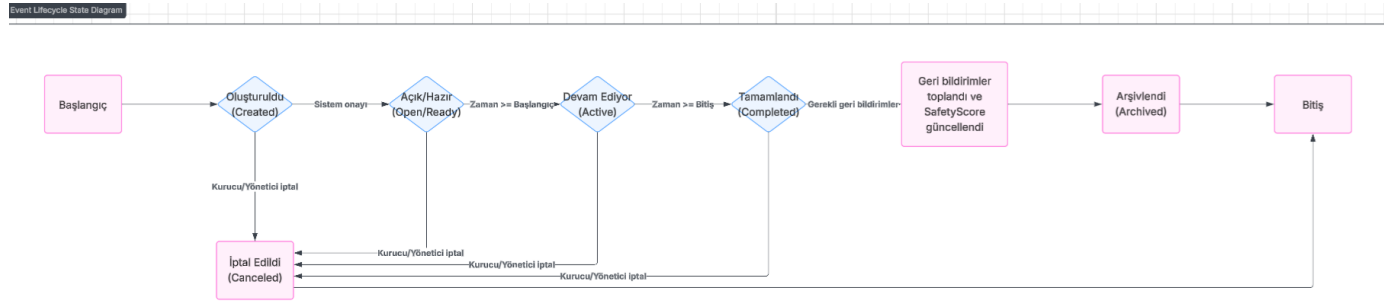


STATE MACHINES

Event Lifecycle Diagram

This diagram presents the Modular Monolith Architecture running within the Application Server Node (Node.js/Express.js Runtime). The Authentication Module serves as the security gateway, directing verified traffic toward key services. The central component is the Matching Engine (Core Logic), which executes the recommendation algorithm based on crucial inputs received from the Event Management Module, the Real-Time Availability Module, and the User Management & Feedback Module. This internal structure ensures all features, including the Chat Module (WebSocket Server), utilize a common, organized set of services to maintain performance and data consistency within a single deployment environment.



User Authentication Diagram

This sequence diagram illustrates the User Authentication process, focusing on the steps required to establish a secure session. The User initiates the flow by entering their credentials into the Mobile App. The app then sends a POST /api/auth/login request, typically containing a hashed password, to the Backend Server. The server verifies these credentials against the Database. If successful, the Backend Server sends a 200 OK response back to the Mobile App containing a JSON Web Token (JWT), which the app uses for secure subsequent requests. Finally, the Mobile App uses the granted access to Display the Main Menu to the User.

